KENYA ELECTORAL PARTICIPATION ANALYTICS

Project Brief

An end-to-end analytics study of how voter registration and electoral participation changed across Kenya's 2013, 2017 and 2022 General Elections.

Status	Project definition
Current phase	Project brief
Next phase	Data acquisition and source assessment
Primary geography	47 counties
Election cycles	2013, 2017, 2022

1. Project Overview

Kenya's registered electorate has changed substantially across recent election cycles. However, growth in registered voters does not necessarily mean equivalent growth in electoral participation.

This project will use publicly available Kenyan election and demographic data to explore how voter registration and participation changed across the 2013, 2017 and 2022 General Elections.

The project will be developed as an end-to-end data analytics case study covering data acquisition, validation, cleaning, modelling, exploratory analysis, statistical analysis, visualisation and communication of findings.

The analysis will remain politically neutral and focus on electoral participation rather than evaluating political parties or candidates.

2. Primary Research Question

> **How has Kenya's electorate changed between 2013 and 2022, and what can the data tell us about voter registration and participation across different parts of the country?**

3. Supporting Questions

- **1.** How did the number of registered voters change between 2013, 2017 and 2022?
- **2.** Did growth in registered voters translate into similar changes in voter participation?
- **3.** Which counties experienced the largest increases or decreases in voter turnout?
- **4.** How does voter registration compare with the adult/voting-age population across counties?
- **5.** Are there geographical differences in registration and participation?
- **6.** How does the number of registered voters per polling station vary across Kenya?
- **7.** Is there an observable relationship between polling-station load and voter turnout?
- **8.** What additional patterns emerge from the data that were not part of the original hypotheses?

The final question is intentional. The project should allow the evidence to generate new analytical questions rather than forcing the data to confirm our initial expectations.

4. Initial Scope

Election periods

- 2013 General Election
- 2017 General Election
- 2022 General Election

Geographic levels

The initial analysis will focus on Kenya's **47 counties** to maximise comparability across election cycles. Where reliable data permits, the analysis may subsequently extend to constituencies, wards and polling stations.

Primary analytical focus

- Registered voters
- Votes cast
- Valid votes

- Rejected ballots
- Voter turnout
- Registered-voter growth
- Turnout change
- Polling-station distribution
- Registered voters per polling station
- Demographic context

Candidate-level results may be retained in the underlying dataset where useful but are not the primary focus of the project.

5. Proposed Data Sources

Electoral data

Primary source: Independent Electoral and Boundaries Commission (IEBC)

Potential supplementary/structured sources: Kenya Gazette election records, ELOG, CitizenGuide Kenya, VotePulse and Kura254.

Secondary datasets will be validated against official published totals wherever possible.

Demographic data

Primary source: Kenya National Bureau of Statistics (KNBS)

- 2019 Kenya Population and Housing Census
- Population by county
- Population by age
- Population by sex
- Urban/rural population
- Population density

Where demographic data and election data refer to different dates, this will be clearly documented and reflected in interpretation.

6. Proposed Technology Stack

Technology	Primary use
Python	Acquisition, profiling, cleaning, transformation, validation, EDA, statistical analysis and feature engineering
SQL	Structured storage, modelling, analytical queries, aggregation and reproducible transformations
Power BI	Data modelling, DAX measures, interactive analysis, geographic visualisation and final dashboard
GitHub	Version control, documentation, code, methodology, data dictionary, outputs and portfolio presentation

7. Proposed Data Pipeline

Source → Raw → Validated → Cleaned → Modelled → Analysed → Visualised

Original source files will be retained unchanged. Transformations will be reproducible through code rather than manual editing of source files.

8. Data Validation Principles

- Record counts
- Duplicate records
- Missing values
- Geographic identifiers
- County naming consistency
- Data types
- Impossible values
- Turnout calculations
- Election totals
- Registered-voter totals
- Votes cast
- Valid and rejected votes

Where possible, **aggregated county results should reconcile with published national totals.** Any unresolved discrepancies will be documented rather than silently corrected.

9. Initial Hypotheses

Hypothesis 1

Growth in registered voters does not necessarily correspond with growth in voter turnout.

Hypothesis 2

Changes in electoral participation differ substantially between counties.

Hypothesis 3

Polling-station load may be associated with voter turnout.

These hypotheses are questions to test, not conclusions the project is expected to prove.

10. Proposed Analytical Measures

Measure	Definition
Turnout Rate	Votes Cast ÷ Registered Voters
Registered Voter Growth	(Current Registered Voters - Previous Registered Voters) ÷ Previous Registered Voters
Turnout Change	Current Election Turnout - Previous Election Turnout
Rejected Ballot Rate	Rejected Ballots ÷ Votes Cast

Measure	Definition
Polling Station Load	Registered Voters ÷ Number of Polling Stations

Additional measures will be introduced as the analysis develops.

11. Expected Outputs

- **1.** Clean and documented analytical datasets
- **2.** Python notebooks/scripts
- **3.** SQL database and analytical queries
- **4.** Data dictionary
- **5.** Methodology documentation
- **6.** Data-quality and reconciliation report
- **7.** Exploratory and statistical analysis
- **8.** Interactive Power BI report
- **9.** GitHub project repository
- **10.** Final project README
- **11.** LinkedIn case-study article
- **12.** Supporting visualisations/screenshots

12. Power BI Report Concept

Page 1 - Kenya's Electoral Story

National trends in registered voters and participation across election cycles.

Page 2 - Geography of Turnout

County-level turnout and changes over time.

Page 3 - Registration vs Participation

Comparison between registration levels and actual participation.

Page 4 - Electoral Infrastructure

Polling stations, registered voters per station and their relationship with participation.

Page 5 - County Explorer

Interactive exploration of individual counties across election cycles.

A separate **Methodology & Definitions** section will document sources, measures, assumptions and limitations.

13. Analytical Principles

Evidence before conclusions. Hypotheses will be tested rather than confirmed.

Correlation is not causation. Relationships observed in the data will not automatically be interpreted as causal.

Definitions before calculations. Every important KPI will have a documented definition.

Validation before visualisation. Data will be reconciled before being used in the final dashboard.

Transparency about limitations. Missing data, inconsistent sources and methodological limitations will be documented.

Reproducibility. Where practical, another analyst should be able to reproduce the analysis from the documented sources and code.

14. Success Criteria

- Reproduce reliable historical election statistics from documented sources.
- Explain how Kenya's registered electorate and participation changed between election cycles.
- Identify meaningful geographical patterns without partisan interpretation.
- Demonstrate professional data-quality and validation practices.
- Demonstrate Python, SQL, data modelling, statistical analysis, Power BI and analytical storytelling.
- Produce findings that can be communicated clearly to both technical and non-technical audiences.
- Be reproduced and understood by another analyst using the GitHub repository.

15. Project Status

Current phase	Project definition
Next phase	Data acquisition and source assessment
Analytical conclusions	None at this stage